MALMÖ UNIVERSITY
FACULTY OF CULTURE
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Digital nudges

Bringing awareness of excessive consumption of digital content in smartphone users

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Abstract

The way we use interactive mobile technology everyday is changing the human cognitive landscape and how we are dealing with reality. In our daily life, SnT (Social-networking technology) has become something we cannot avoid using, as the need of being constantly updated and virtually connected to our social and working circle has prominently emerged. In this Interaction design thesis, I intend to explore the possible ways in which digital users, physical people in the first place, can prevent addictive loops of being constantly connected to such services and smartphones. Ethnographic research methods were performed along with iterative prototyping phases, leading to the development of speculative design interventions. Altering different responsiveness rates of a mobile user experience led me to design six bodystorming concepts and enact relative roleplay activities supposed to prevent SnT addiction and eventually help people being in control again of their spent digital time.

Table of Contents

Abstract **2**
Table of contents **3**
1. Introduction **4**
1.1 State of the field 4
1.2 Research focus 5
1.3 Background definition 5
2. Model and theory **9**
2.1 Related work 11
2.1.1 Focus keeper 11
2.1.2 Forest app 12
2.1.3 One sec 13
2.1.4 iOS Screen time 14
2.1.5 Android digital wellbeing 15
2.1.6 Gripsense and Vibpress 17
3. Methodology **17**
3.1 Double diamond 18
3.2 Literature review 18
3.3 Semi-structured interviews 19
3.4 Affinity diagramming 19
3.5 How-might-we 19
3.6 Questionnaire 20
3.7 Sketching and prototyping 20
3.8 User testing 21
4. Design process **22**
4.1 Discovery phase 23
4.1.1 Literature reviews 23
4.1.2 Semi-structured interviews 23
4.1.3 Questionnaire 23
4.2 Define phase 24
4.2.1 Affinity diagrams 25
4.2.2 Early prototyping 26
4.2.3 How-might-we 26
4.3 Iteration 27
4.3.1 Preliminary user testing 27
4.4 Develop phase 28

4.4.1 Sketching 28
4.5 Deliver phase 29
4.5.1 Bodystorming concepts 29
5. Discussion 33
6. Conclusion 34
Acknowledgments 36
References 37
Appendix 39

1. Introduction

Communication has a vital role in our everyday-life. Mobile technologies have gradually become an increasingly important communication tool in both private and professional activities, along with web access, nowadays available from almost every technological device. Teenagers and young adults represent a consistent part of the actors involved in this great human digital landscape. According to developmental psychotherapists, mobile computation devices seem to greatly influence the psychosocial evolution of teenagers. Teenagers and young adults, humans in general, have the fundamental need to socialize and this is nowadays enhanced or mediated by social networking, and research in this field has shown that the well-being of adolescents is strongly influenced by their use of social media. Social media sites have rapidly become a central part of young people's lives, with over 90% now using social media, day and night (Duggan & Smith, 2013).

Social media habits represent one of the most common – and controversial – debated questions of habitual behavior in contemporary society. In this thesis project, an overview of the state of research on social media usage and related digital wellbeing aiding tools is summarized, along with their relative position within literature.
In particular, night-time usage of digital devices and SnT services affects the overall sleep quality patterns of humans. It has become evident how nighttime-specific use and emotional investment in social media relates to sleep quality, self-esteem, anxiety and depression in adolescents (Woods and Scott, 2016).

Methods, data gathering activities and design results were analyzed, performed, and proposed in this research process. The first phase of this analysis relies on the observation: semi-structured interviews and a questionnaire helped me to understand how much emotional awareness there is in users experiencing SnT and digital control tools. In the following chapter the contextual details and practical outlines of the interaction design project are presented. The last phase of the research involves an analysis on how changing the user experience of a digital artifact would affect its perception of use. Consequently, results are shaped into roleplaying speculative concepts supposed to help users disconnect from digital devices. As a whole, this thesis demonstrates the need to break apart the addictive components of social media habits in order to clarify their implications for human well-being.

1.1 State of the field

This thesis and the following research process investigate the concept of an overconnected "Networking society" and our perception of it through an Interaction Design perspective. A deep understanding of human psychology is essential when creating user-centered products with proper user experience. Developing an understanding of human cognitive limitations and capabilities is the key of interactive designs. Without an awareness of how we interact with things in real and virtual worlds, it is hard to design valuable interactive user experiences. The psychological knowledge of how we consider users, shifting through different aspects of human cognition, and learning how these relate to interactions and consequent social behavior, is a necessary premise to create user-centered products, free from unwanted or uncontrolled consequences.

Based on speculative design approaches (Mitrović, Hanna & Helgason, 2021), the design research outcome will focus on re-thinking the experience of SnT usage, proposing an exploratory knowledge contribution approach while creating concepts for enabling users to gain control over the spent digital time on applications.

I will investigate on a rising phenomenon called “Fear of Missing Out” (FOMO) as well, considered as a general state of anxiety at missing out on rewarding experiences, such as smartphone notifications, often driving social media engagement (Przybylski, Murayama, DeHaan, & Gladwell, 2013). Young people in particular report feelings of disconnectedness and missing out without access to online communication, preferring to keep their phones within reach at night (Vorderer et al., 2016). This desire to be constantly connected and concern about missing out when offline may make it difficult for highly invested adolescent users to disengage from social media at bedtime (Woods & Scott, 2016).

1.2 Research focus

The current research explores the idea of altering the user experience of a SnT service mobile session (e.g. changing the touch responsiveness, altering colors, adding response delays, changing interactive modalities, etc.) once the preset daily time limit of use has been reached. The aim of the study is to better understand the psychological outcomes of addictive digital user experiences, along with speculating on a design proposal giving users feedback about the excessive time they might be spending on mobile technology, providing a feature that progressively reduces accessibility and consequently addictiveness.

The main research question driving the design process is:
“How can the aesthetics of interaction be used to give feedback on time spent consuming social media content?”. The following chapters of the thesis project are aimed at answering this question.

1.3 Background definition

In this section, the current scenario of intervention is outlined, to define a specific design opportunity to intervene on. Before going through the design process’ phases, it is necessary to extract from the research question the area of intervention of the problematic scenario.

Modern computing and ”Information and communications technologies” (ICT) are the infrastructure and components of our everyday digital devices’ scenario, composed of computers, audio-video technologies and related software and services, which allow users to create, store and exchange information. It has drastically changed how people work, communicate, learn and live. The development of the technology of Web 2.0 has been the revolution in the concept of "Managing your own Internet", where users were able to upload personal content and directly interact with other users.

Therefore, in terms of dealing with other people, the barriers of facing physical frontal interaction have disappeared, in favor of experimenting new ways of coming together. ”Social media” are defined as ”virtual platforms that enable the creation, publication and sharing of user-generated content”. There are many benefits if such web services are properly used, but overusing them in our current “online era”, in a continuous blend between real and virtual worlds, may entail personal health risks.

For example, a misalignment between biological time and time spent on social media causes "social jet lag", impacting our sleep processes and neurobehavioral functioning. Examining the interplay between biological rhythms and technology-mediated social interactions, we find that technology and social connectivity may modulate and misalign circadian rhythms and body clocks. (Murnane et al. 2015)

According to the "Global digital overview" report (fig.1), the connected world continues to grow faster: 5,31 billion unique mobile phone users were estimated by the start of 2022, more than two-thirds (67.1 %) of the world's population (7,1 billion). Global internet users have climbed to 4.95 billion at the start of 2022 and 4.62 billion social media users around the world in January 2022. This figure is equal to 58.4 % of the world's total population. Global social media users have grown by more than 10 percent over the past 12 months, with 424 million new users starting their social media journey during 2021 (fig. 2). The advantages of internet communication are many, such as a possible access to a wider network of people with similar interests or concerns and increased ability to stay in touch with geographically distant friends and family.

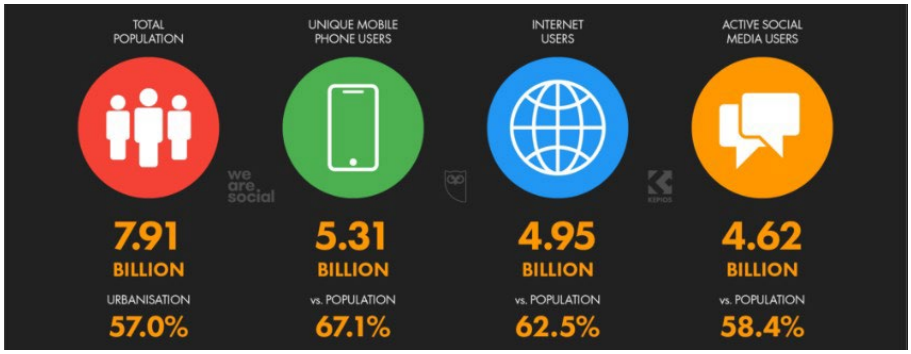


Figure 1 - Global Digital Overview - January 2022 Source: DataReportal

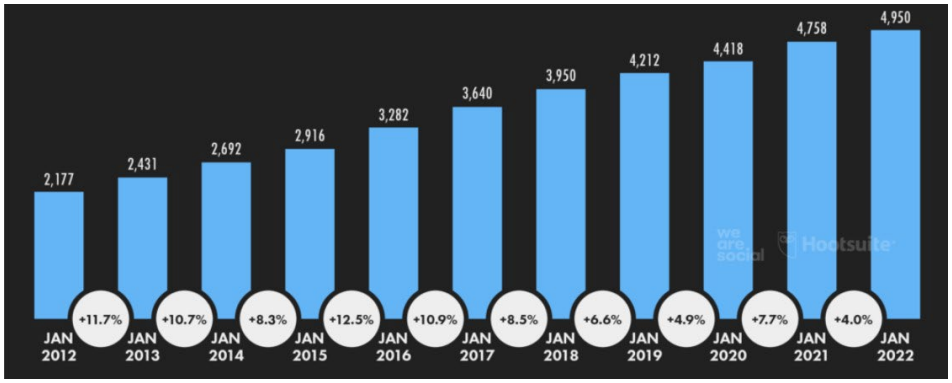


Figure 2 - Internet users over time. Source: DataReportal

Researchers from different disciplines are constantly keeping up with the dynamic growth of internet usage, trying to understand the different consequences of technology innovation. In the case of social media this is done through the analysis of users, their motivations of use, their way of building an online identity, the ways of interacting online and the related culture, as well as the real-life social consequences, and the constant rise of privacy-related issues. The social benefits of using phones and social media are unrivaled in human history: from the rapidity and ease of connecting with friends, to keeping up to date with relevant cultural information.

However, researchers described some of the main risks facing new ”network sociality”, picturing the possibility of an improper usage of networked technologies amongst users. The many downsides of an uncontrolled social media usage could be states of anxiety, low self-esteem, bad sleep quality and depression (Woods and Scott, 2016), as well as might involve the feeling of being isolated. Digital users perceive on-line interaction and the usage of social networking services as less threatening than face-to-face interaction and believe themselves to be more efficacious in their social interaction in this context (Caplan, 2006). During Covid-19 pandemic and related lockdown measures taken by governments of many countries have caused a rising use of digital devices and social media. Studies revealed that feelings of loneliness had a higher negative impact on users' happiness than feelings of anxiety and acknowledged the role of social media as a constructive coping strategy for users to deal with anxious and loneliness feelings (Cauberghe et al, 2021).
Statistical and medical researches made in recent years found out that social media networking services are addictive (Macit et al 2018). Research has been conducted on changes in chemical structures and physiological behavior of individuals' brain and nervous system. Internet addiction is compared to other types of behavior and symptoms of real substance abuse causing biological and psychological addiction like alcohol, cigarettes, pills and chemical drugs, which affect the physiological mechanism of experiencing pleasure, in particular the dopamine release. In this context, there are potential risks to affect human life in a negative way and create unwanted addictions through inappropriate use of the Internet.

Although positive uses of the internet are numerous, awareness of the potential disadvantages of its communication ways are growing. Kraut et al. (1998) examined the effect of internet usage and speculated that, even if most internet uses are devoted to active social communication, it might end up reducing face-to-face contact, to the point that weaker social relationships replace stronger ones. Internet use may serve as a way to avoid being alone and may intensify disconnection from real-life relationships. Electronic devices, thanks to their immersive user experience, may alienate ourselves from reality and lead us to cope with new ways of being alone (Turkle 2008). Being always online and connected inevitably leads to an overload of information and stimuli, negatively enhanced by a seamless everyday stream of digital notifications weakening personal attention and concentration. This digital behavioral pattern has gradually weakened the ability to bond meaningful relationships, placing more value on their external perception, leading to the eventual generation of negative overusage symptoms.
A research conducted by the scientific observatory of a non-profit organization demonstrates that 79% of teenagers between 11 and 18 years-old spend more than four hours a day on social media: two whole months in a year. Any attempts of interruption gives them typical attitudes of a possible addiction. 52% of the study participants have unsuccessfully attempted to reduce their online time and 33% of them are self-aware of excessive smartphone consumption behaviors. The most evident consequence is irregular sleeping patterns; 40% of respondents claim to lose hours of sleep because they remain connected and active online at night, reducing well-being and productivity.

Studies on these issues lead to the assumption that the effects of social media on mental health are not so much related to the social networks themselves but to the distorted use we may end up making of them (Morgiève 2019). According to the scientist, there are three main types of distorted use of social networks: the first is related to social comparison, the spontaneous tendency of social beings to compare themselves with individuals who appear to possess a stronger attractiveness; the second problem is their propensity to promote addictive behavior,

including the fear of missing out on important news. The main symptoms of developing an addictive condition are: the increasing feeling of absolute necessity to stay connected, for gradually longer periods of time, to achieve the same level of pleasure (tolerance); when prohibited from using social networks, people typically show signs of impatience, discomfort or irritability for the mere fact of not being able to access their social accounts (abstinence); entertaining virtual relationships becomes the primary concern and conflicts with any other activity (the conflict). In addition, a new phenomenon called 'Fear of Missing Out' (FoMO) has been outlined, consisting in the fear of 'not being there', or being cut off from events organized by friends if not connected to social networks.

Social media habits are recognized by researchers as one of the most common forms of habitual behavior being created in contemporary society (Joseph B. Et al., 2022). Internet-related services of media and content production, could be framed in this way as habit-making means. In his research, Schulz (2013) stated that "Rewards produce learning", meaning that social media usage, as a result, ends up forming a habit. Internet users are compared to Skinner's rats learning to press levers to attain rewards or Pavlov's dogs drooling at the sound of a bell. They both learned to associate that behavior to a reward. As well as Skinner's rats and Pavlov's dogs, social media generates a multitude of rewards and stimuli that result in learning and habit-forming (Monaco, 2020). This habit formation is mediated by dopamine, a neurotransmitter, in brain areas known to be involved in reward processing. Despite common knowledge, dopamine release is not directly correlated to delivery of reward; instead, recent research has shown it is involved in the prediction of reward (Schultz et al, 1997). This can be applied to the digital feature of receiving 'likes' on social media. Notifications play a crucial role: popping up on the phone at unpredictable times, they act as cues for reward, leading to positive dopamine efflux. Through this repeated reinforcement learning we form habitual behavior where we constantly check our phones for more rewards.
Social Network Sites (SNSs) are meant to facilitate interaction between people. The design of SNSs employs persuasive techniques with the aim of enhancing the user experience but also increasing interaction and user retention (Andreassen C., S.). In the case of social media apps, the relative addiction consists in the excessive and compulsive need of online social networking.

Social media addiction concerns have increased steadily over the past decade. Digital nudges have been shown to hold enormous potential to change behavior. However, it is not clear how they might be designed to combat social media addiction (Purohit et al 2020).
Digital Nudging denotes all those features of virtual environments that influence people's behavior in a predictable way. Nudges are known to be a key principle to understand and direct the process by which people make decisions. Used appropriately, nudges can guide people to make better choices. The same design principles that create addiction could be leveraged to mitigate it. More specifically, digital nudges, i.e. design features that steer people's decisions without banishing freedom of choice, can be used in welfare-promoting directions and can be used by designers to mitigate the addictive features of social media (Purohit et al 2020).
To understand the design strategies causing addiction in the use of a certain digital service, in this interaction design thesis I will later refer to the "Hooked model" (Nir Eyal 2014), which structures the addictive use of digital services in four main phases: Trigger, Action, Reward and Investment. This contextual viewpoint depicts marketing strategies behind the design of these digital services as aiming to create a behavioral habit in their usage.
In addition, another theory behind the design of habit-forming products and services, is the uses and gratifications theory. Gratifications are conceptualized as the satisfaction people receive by using particular media that meet their expectations. If a particular media satisfies the user's needs, that will result in the recurring usage of that media (Menon 2021).

This interaction design thesis project aims to create new speculative design proposals to disengage from social media mobile interfaces, improving awareness of the time spent in consuming user experiences of digital devices. Once examined the theoretical framework, the necessity of providing a solution to the digital addiction has been raised. As a matter of fact, preventing digital addiction would lead to solving related problematic outcomes of irresponsible uses of such digital tools and services, eventually causing social issues.

2. Model and theory

This section reports the result of the main literature review, including theories and models that make up the theoretical approach of the thesis, investigaing possible user experiences enhancing well-being, trying to solve addictive usage patterns.

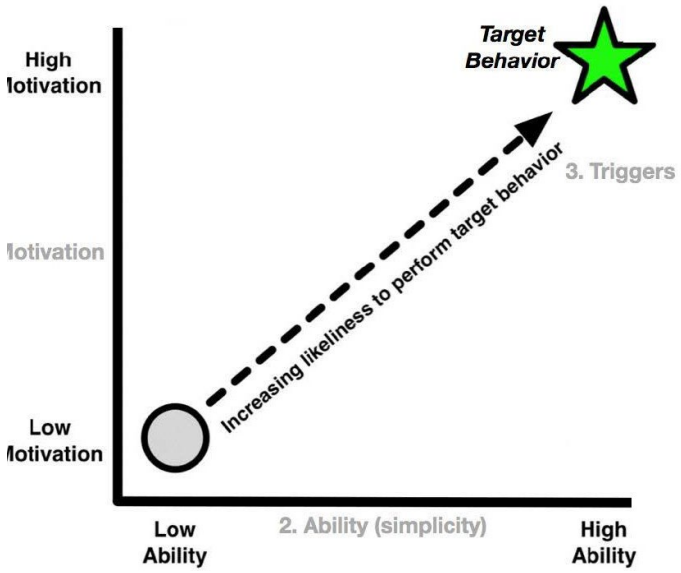


Figure 3 - The Fogg Behavior Model (BJ Fogg, 2009)

Therefore, focusing on the application of the model in Interaction design, the attention is shifted on the possibility that designers create products and services according to different emotional states of the users and their willingness to perform certain types of actions. According to BJ Fogg's model (fig.3), for a target behavior to happen, a person must have sufficient motivation, sufficient ability, and an effective trigger. All three factors must be present at the same moment for the behavior to occur (BJ Fogg, 2009).
BJ Fogg argues that persuasive technology is fundamentally about learning to automate behavior change. On the other hand, if designing products for serious tasks, such as driving a car, designers must pay attention to all the information needed to complete the task at hand and that the interfaces must be visible with clear and unambiguous feedback. Another application of the model is designing products in terms of the three levels: visceral, behavioral and reflective design of product. According to these three paradigms, a product will be designed

using aesthetics techniques to make the product emotional using particular color, shapes or properties known to evoke certain specific emotion.

Purohit et al. (2020) argue that social media platforms experiences are designed in an engaging way, leading to undesirable platform usage that can bring to addictive usage patterns. The Hooked model (fig.4) describes the design of the development of habit-forming products. It is composed of a four-phase iterative process starting with a trigger that leads to an action, which produces a reward and ultimately creates an investment. These four phases produce an addictive feedback loop where the more users receive rewards, the more they will be incentivized to respond to triggers and perform actions, producing more rewards and investments.

When sufficient ability and motivation exist with a trigger, an action occurs. Actions in social media are typically browsing, posting, commenting, or reacting. Social media platforms increase the ability of users to take actions by making actions more accessible. The motivation to seek pleasure, social acceptance, and to avoid pain and social rejection increases the chance that the user will take action. Giving users rewards after they have taken action brings them to social media platforms over and over again. Rewards can be of three types: the 'tribe' (comments and likes), the 'hunt' (new content consumption and endless news feeds), and the 'self' (intrinsic rewards of mastery, competence, and completion). Investment increases the likelihood of users returning to social media platforms as they have invested value in the form of content, data, followers, and reputation. The investment on the platform increases the possibility of users responding to the next trigger to start the cycle once again.

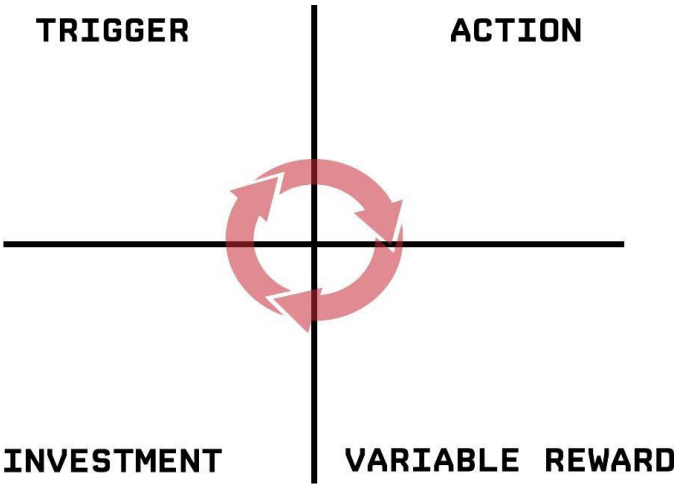


Figure 4 - Hooked model (adapted from Nir Eyal, 2014)

Another interesting model that inspired my project reflection has been the one describing the relationship between the fear of missing out on digital devices and the night-time usage of the same devices. It has been theorized how FOMO predicts nighttime social media use (fig. 5), which is in turn linked to shorter sleep duration via two paths: theoretical framework of electronic media use and sleep in children and adolescents (Cain et al. 2010). It has been proposed that nighttime social media use will predict later bedtimes, in line with similar findings in young adults (Orzech et al. 2016). Adolescents who used social media more around bedtime tended to take longer to fall asleep, and this was mediated by how alert they felt in bed

(Harbard et al. 2016). By delaying sleep onset, both these possible mechanisms are particularly relevant for the sleep duration of adolescents, whose school timetables prevent later rise times to compensate (Owens et al, 2014). Scoots and Woods (2018) analyzed how FOMO predicts nighttime social media use, which is in turn linked to shorter sleep duration on a behavioral level (via later bedtimes) and cognitive paths (via increased pre-sleep cognitive arousal and longer sleep latency).

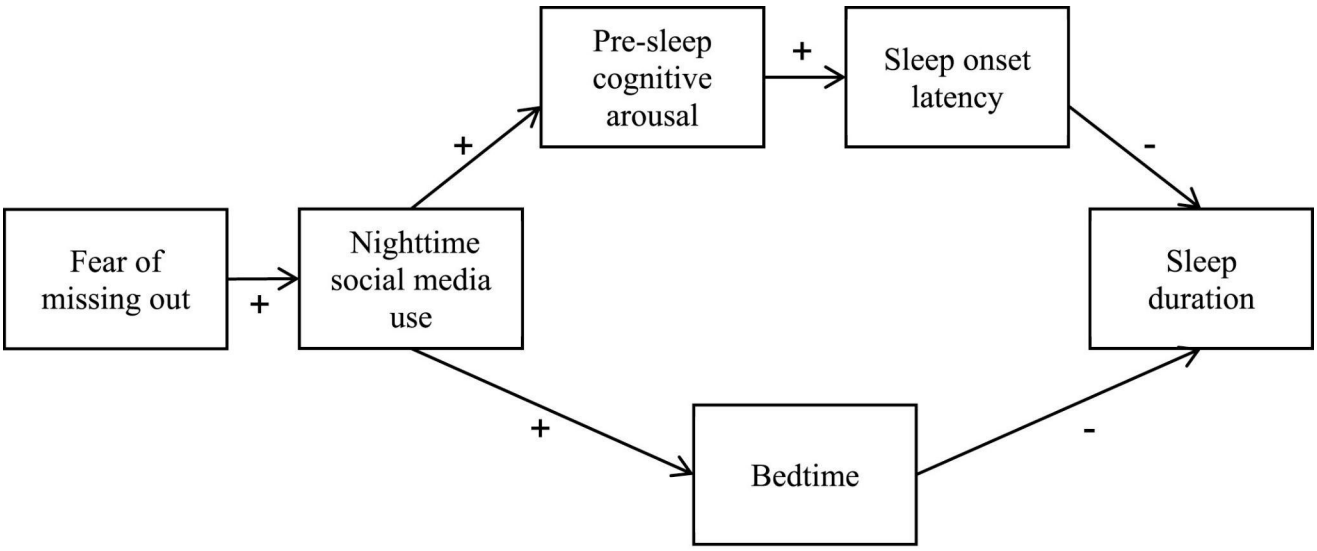


Figure 5 - FOMO and related phenomena model

2.1 Related work

The following case studies represent available mobile applications and software helping users to disconnect or be in control of their time spent on digital devices. Several services and features were analyzed in order to possibly isolate the most promising aspects to be taken as further inspiration, in the creative process, to work on and possibly innovate this particular subfield of Interaction design.

2.1.1 Focus keeper

Focus Keeper is a customizable timer and task management tool working on desktop and mobile apps. This app (fig. 6) is inspired by Pomodoro Technique which is a time management method. It helps users stay focused on any task they may be working on, such as studying, writing, performing activities or getting things done. Using its timer, it is designed to help users keep their productivity high, avoiding distractions or burnouts.

The user chooses a single topic to focus on, spending 25 minutes working on that topic while avoiding multitasking. Once the time has passed, he can take a short break. After every four cycles, he takes a long break. The cycle goes on as the user prefers, with the possibility to choose the breaks and focuses phases' length.

The strategy behind this application is to give an incentive of auto-control to the same users. By starting a timer, they are incited not to engage with their smartphone and instead focus on their tasks and duties to be carried out.



Figure 6 - Focus keeper (focuskeeper.co)

2.1.2 Forest app

Forest is a mobile application (fig. 7) that fights your need for mobile technology, helping you stay focused on the important things in your daily life. Whenever the user wants to stay focused, he can digitally plant a tree on the app. The tree will grow while the user focuses on his own work. Leaving the app halfway, to use the phone differently, will cause the tree to die. Growing a digital forest of trees will help users stay focused daily and turn their hard work into a visual land of lush forest. The concept behind it is to stay focused avoiding digital distractions, while working at the office, studying at the library, together with friends. Forest team partners with a real-tree-planting organization, Trees for the Future, to plant real trees on Earth.

This app has the same strategy as the Focus Keeper application, to give an incentive of auto-control to the same users. By starting a timer, they are incited not to engage with their smartphone and instead focus on their tasks and duties to be carried out. The visual representation of the growing plant - which would eventually die if the app would be quitted - gives the user another reason not to engage with the smartphone.

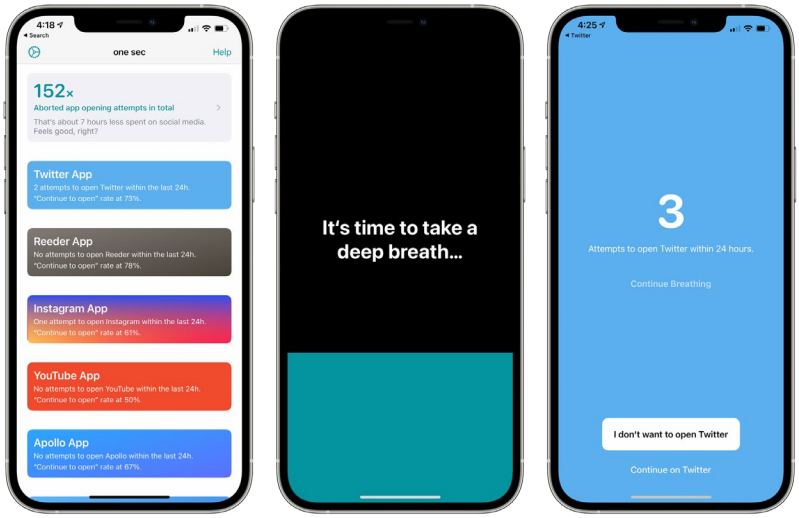


Figure 7 - Forest app (forestapp.cc)

2.1.3 One sec

One sec (fig. 8, 9) is a mobile application that delays distracting apps and websites of users' choice: it forces the user to take a deep breath whenever a social media app is open. The way it is designed is as simple as effective: additional friction removes instant gratification and makes distracting apps less appealing. It helps users gain back control over their spent digital time.
Social media apps, in particular, are designed to keep users hooked to spend as much time as possible on them. Time that could be spent in a better way, with family and friends, getting work done, or for hobbies and wellbeing. Our social media habits are the outcome of psychological mechanisms of social needs that apps trigger in our brains. They use the system of instant gratification, which releases dopamine rushes whenever we perceive a like or a message on our phone. By making the user wait longer for social media to open, the brain loses interest in those (destructive) short-term dopamine hits. Many apps and tools block social media for a limited amount of time to help users focus, and this might increase users' productivity for the moment, but the real problem – the habitual urge to check apps and websites to get that dopamine boost – stays.

The strategy behind the app is to set up concrete, healthy alternatives to help the user get away from social media habits: looking at family photos, doing sports, calling a friend, drinking water, reading a book, or listening to music. Alternatives will be suggested everytime users try to open an app or website they want to control the access of. Users will be forced to take a deep breath and then decide if they really want to use it, controlling boredom or stress impulses. They are incited to act intentionally: if configured, one sec forces the user to set an intention for opening apps. The app also gives the opportunity to visualize the progress in nice diagrams and comparisons.



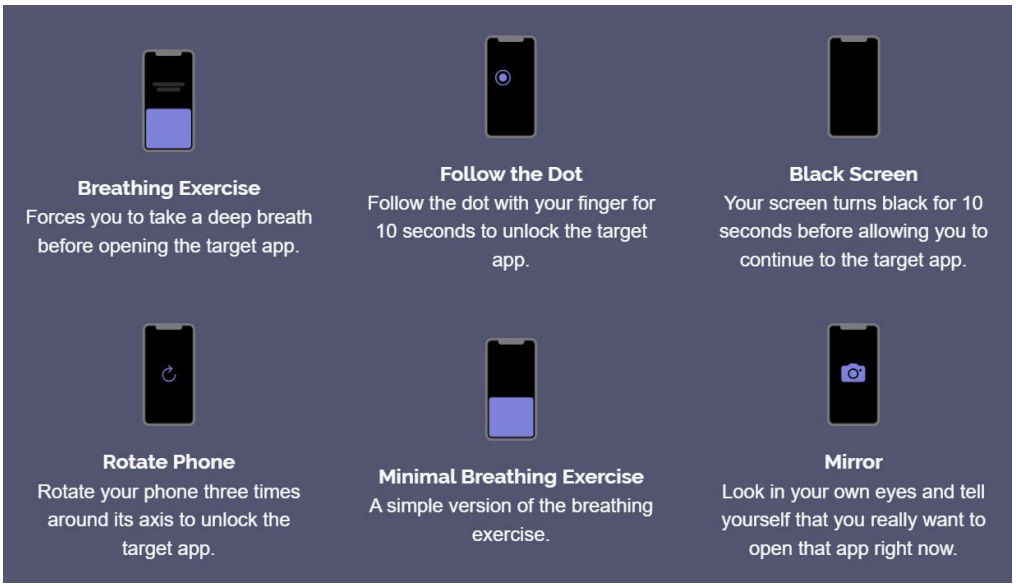


Figure 8 and 9 - Onesec app (one-sec.app)

2.1.4 iOS Screen time

Apple's Screen Time service (fig. 10) aims to address growing concerns around increasing device usage, smartphone addiction and social media impacting on mental health. It helps users to monitor the way they use devices, presenting everything from how much time has been spent on certain app categories and specific apps, to how many times users have picked up their iPhones in a given day or even hour. By understanding how users interact with their phones or tablets, they can then take much greater control, either by taking drastic action like removing the app completely, or by limiting its use by setting App Limits.

However, users have got to decide to make that change. They can choose to ignore the Screen Time findings or bypass them at any time. When an App Limit is about to expire, the user will get a notification saying that only five minutes are left. When time is up, the app icon will be slightly grayed out on the home screen and if opened, a white screen with an hourglass and the words 'Time Limit' will appear. Users can choose 'OK' to leave the app and do something else with their time. Or, they can select 'Ignore Limit', after which they'll have four choices to select from: 'One more minute', 'Remind Me in 15 Minutes', 'Ignore Limit For Today' or 'Cancel'. Users can choose to dismiss it, but it does make them think twice.

Downtime is another feature within Apple's Screen Time that allows users to schedule a block of time in which only certain apps users choose will work. It's possible to whitelist certain apps so users can still use phone or messaging features. The strategies behind the features of this app are to enhance the users' self-control and awareness of the digital time spent.

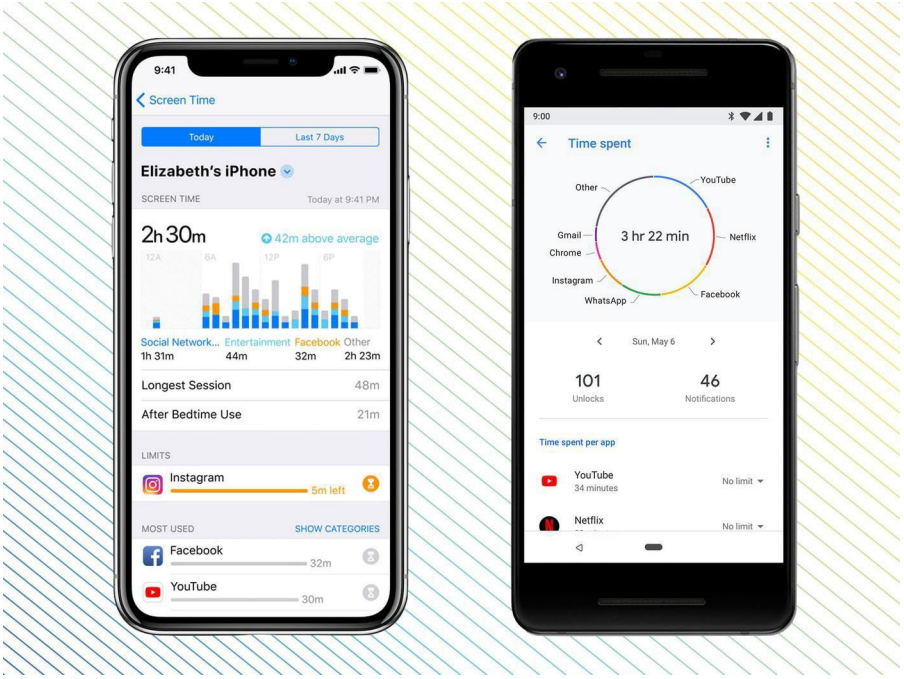


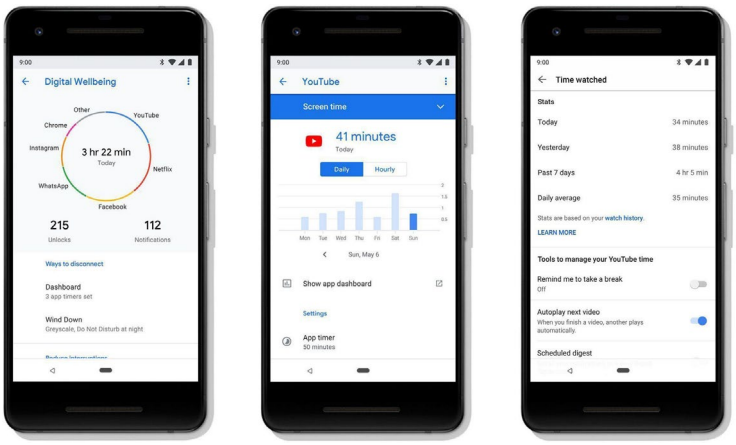
Figure 10 - iOS Screen time (support.apple.com)

2.1.5 Android digital wellbeing

Digital Wellbeing is an Android feature that assists users in understanding and managing their digital behaviors (fig. 11, 12). It aims to encourage a healthy and thoughtful relationship with technology, offering a variety of tools and features to assist users in having a more balanced connection with their digital gadgets. It provides an overview of digital habits, including screen time, times of screen unlocks, and the most used apps. It also offers suggestions for setting limits and taking breaks from the device itself. Users can limit time spent on specific apps and whenever reached that limit, the app itself will be grayed out, while receiving at the same time a notification reminding users to take a break.

A feature called Wind-down helps users prepare for bedtime, by turning on Do Not Disturb mode, fading the screen to grayscale, and hiding all notifications. Removing all colors from the screen, makes the smartphone itself less appealing, and helps the user to reduce uncontrolled screen time behavior.
Recent research findings highlight the importance and strength of individual motivations behind the need to disconnect from mobile technology, framing digital wellbeing not as a tool rather than a subjective consideration of how technology affects users' own lives instead. Understanding personal digital behavior can help put habits into different perspectives, but ultimately, it is on the user's own willingness to act according to that data.

The strategy behind the service is to bring awareness of excessive use and to prevent app overuse, but it may not work efficiently, because even if the individual intends to change, they won't always take action. This inconsistency has been called the "intention-behavior gap", meaning that people do not always do the things that they intend to do (Sheeran and Webb, 2016). Ideally, one way to help bridge that gap could be improving the UX features, implementing a leveling-up system with weekly rewards for meeting goals, as more people might try to achieve them in this way.



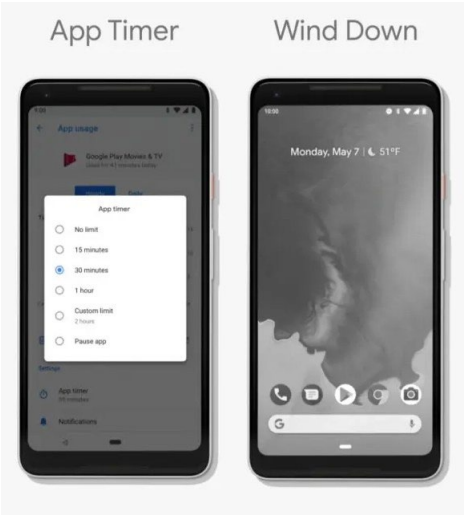


Figure 11 and 12 - Android digital wellbeing (android.com/digital-wellebeing)

2.1.6 Gripsense and Vibpress

Gripsense is a system that leverages various capabilities of mobile devices like the touchscreen, inertial sensors, and the vibration motor to infer users' hand postures and the amount of pressure exerted on the device (fig. 13, 14). It uses smartphones' built-in inertial sensors and vibration motor to infer one or two-handed gesture interactions, using thumb or index fingers. It also senses the amount of pressure a user exerts on the touchscreen despite a lack of direct pressure sensors by observing diminished gyroscope readings when the vibration motor is "pulsed." A similar project, VibPress, is a software technique enabling pressure input interaction on mobile devices, by measuring the level of vibration absorption with a built-in accelerometer. This is achieved using a real-time estimation algorithm running on the device when the device is in contact with a damping surface (e.g., user's hands).

These projects inspired me to consider the possibility of using the built-in sensors of a smart device as a way to collect data about the way it is physically handled. In an ideal situation crossing the sensed parameters would further help to understand the strength and precision of performing manual gestures.



Figure 13 and 14 - Gripsense and Vibpress app concepts

3. Methodology

This section generally describes the methods used in the design process, including analysis and tests that helped develop the theoretical approach of the thesis, that will later be explained later in the design process and discussion.

3.1 Double diamond

The design process has followed the Double Diamond model (fig. 15), a comprehensive and visual description of the design process. The two diamonds represent the process of exploring an issue more widely (divergent thinking) and then taking focused action (convergent thinking). It is composed of four main phases:
1. Discover. The first phase focuses on gathering knowledge about the field of study. It is an exploratory phase and requires divergent thinking. The discovery phase of this thesis is conducted through literature review, and a contextual overview.
2. Define. The second phase requires a convergent mindset. The material gathered from phase one is analyzed and elaborated, ending with a design challenge. The define phase of this project is conducted through constructing a questionnaire and conducting interviews, in order to have quantitative data to work on to envision a possible design challenge as actionable.
3. Develop. The second diamond refers to divergent thinking, exploring different solutions to the defined design challenge. During the development phase of this project design methods such as sketching and ideating have been adopted.
4. Deliver. The last phase involves testing and speculating on different solutions, rejecting those that probably won't work, while implementing the most reasonable idea(s). In this thesis this phase is represented by prototyping, and user-testing, delivering a design concept.

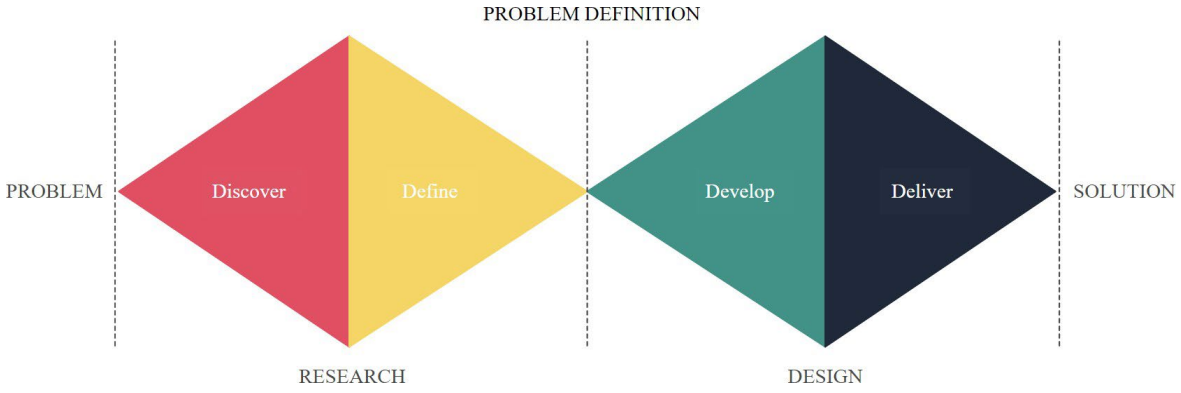


Figure 15 - Double diamond model

3.2 Literature Review

A literature review is a comprehensive summary of previous research on a topic. The literature review surveys scholarly articles, books, and other sources relevant to a particular area of research (Lawrence et al, 2008). It demonstrates knowledge and understanding of academic literature on a particular topic. Reading combined with critical analysis can help to refine a topic and frame research questions. Conducting a literature review deepens understanding of the state of the research in a particular field before carrying out a new investigation. During this phase it is important to summarize and analyze previous research and theories and identify areas of controversy and contested claims.

3.3 Semi-structured interviews

Semi-structured interviews combine features of structured and unstructured interviews and use both closed and open questions. The interviewer has a basic script for guidance so that the same topics are covered with each interviewee. The interviewer starts with preplanned questions and then probes the interviewee to say more until no new relevant information is forthcoming. It is important not to alter an eventual answer by phrasing a question to suggest that a particular answer is expected. The body language of the interviewer, for example whether they are smiling, scowling, looking disapproving, and so forth, can have a strong influence on whether the interviewee will agree with a question, and the interviewee needs to have time to speak and not be rushed.

Martin & Hanington (2012) describe interviews as a primary research method to collect personal expertise, opinions, attitudes, and perceptions. There are different ways to specify the character of an interview, depending for example on the structure, target audience and amount of respondents. An interview may follow a certain formal structure or have a flexible conversational format. In any case, it is recommended for the researcher to have a set of topics to address during an interview. Martin & Hanington (2012) argue that flexible interview format has the advantage of being more comfortable for a respondent, while the fixed script of questions may be seen as impersonal. Yet, an unstructured conversation mostly relies on the researcher's guidance, while the structured set of questions is easier to control and analyze.

3.4 Affinity diagramming

The affinity diagram organizes a large number of ideas. It is the organized output from a brainstorming session. It is used to generate, organize, and consolidate information related to a product, process, complex issue, or problem (Scupin, R., 1997). The affinity diagram process lets a group move beyond its habitual thinking and preconceived categories. In the first phase ideas are placed in random order. Over several days concepts are moved in natural groups of similar items. In the Interaction design field, this technique should be used after gathering information about users through usability testing, surveys, observation, or any other method for collecting feedback. Affinity diagramming is composed of four main steps: grouping insights according to similarities; ranking the clusters according to the scope of the design challenge; exploring connections between clusters; summarizing and investigating gaps in materials (Dam and Siang, 2022).

3.5 How-might-we

How-might-we (HMW) questions are a research technique usually used after the discovery research phase. At the end of a discovery, the team should come together, agree on the main findings, and use this knowledge to frame design challenges. Using this technique at the end of the discovery process can result in framing the design challenge on the right problems. They are statements allowing to reframe insights into design opportunities, to innovate on problems eventually found during user research and to propose efficient design solutions.

3.6 Questionnaire

A questionnaire is a technique for collecting data from respondents about their attitudes, experiences, or opinions. Questionnaires can be used to collect quantitative and/or qualitative information. Questionnaires are similar to interviews in that they can have close or open-ended questions, but it can be distributed to a large number of participants and more data can be collected. Questions should be clearly worded so that the data collected can be analyzed efficiently. Questionnaires can be conducted either online, by phone, on paper or face-to-face.

3.7 Sketching and prototyping

Sketching is an essential moment of the designing process. Sketching and working out multiple possible solutions to the design problem is a way to quickly explore concepts. It is a fast way for rapid prototyping." When we sketch, we begin to see, and that seeing is at the core of design". Through the sketching phase multiple design options are quickly identified. Sketching is a fundamental part of the design process, especially in the early stages. According to Buxton (2011), one of the essential aims of sketching in the ideation stage is "to provide a catalyst to stimulate new and different interpretations".
He argues that rapid and low-quality sketching can truly afford to play, explore, and learn because of the low investment into the process. While "too much concern for quality too early may well have a negative effect". (Buxton, 2011).
Therefore, sketches should be quick, inexpensive, disposable, use minimum details, they should suggest, explore rather than confirm, etc. Buxton emphasizes the importance of ambiguity, arguing that an essential element of sketches is the ability "to be interpreted in different ways". Among other attributes, he also suggests that sketches should "suggest and explore rather than confirm". (Buxton, 2011).

Martin & Hanington (2012) describe prototyping as "a tangible creation of artifacts at various levels of resolution, for development and testing of ideas". Houde & Hill (1997) define a prototype as "any representation of a design idea, regardless of medium". Authors emphasize that regardless of tools and media used to create prototypes, the most important is "how they are used by a designer to explore or demonstrate some aspect of the future artifact".
Houde & Hill (1997) argue that a problem when speaking of prototypes in the HCI field is the complexity of interactive systems and lack of common vocabulary across disciplines. They suggest that designers may focus the exploration on some particular features of a prototype. Authors suggest that by understanding what design questions should be answered, designers may define what kind of prototype to build. (Houde & Hill, 1997).

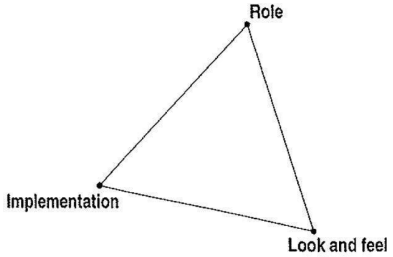


Figure 14 - Houde and Hill model

Therefore, Houde & Hill propose a model (fig. 14) to describe three important aspects of interactive artifacts. Authors define these three aspects as role, look and feel, and implementation. Each dimension aims to answers a set of questions, for example: Implementation usually requires a working system to be built; look and feel requires the concrete user experience to be simulated or actually created;

3.8 User testing

"Usability testing is an evaluative method that allows teams to observe an individual's experience with a digital application as he or she walks through the steps of a given task (or set of tasks)" (Hanington and Martin, 2012). Usability testing in this project focused on what should have been included in the first concept, rather than how it should have worked and was conducted to determine what kind of experience the end-user would have liked to have. This helped the research process to refine all the findings as well as structure things with a hierarchy, based on users' feedback. Usability testing also showed that while some functions and information were desired in the questionnaire and interviews phases, some of these seemed unnecessary when it came to testing out prototypes.

"Prototyping is the tangible creation of artifacts at various levels of resolution, for development and testing of ideas within design teams and with clients and users" (Hanington and Martin, 2012). The low level of fidelity of the first prototype helped the different research eventualities to stay open to the possible iterations and allowed the participants to be more open in giving feedback.

4. Design process

The design process has been organized following the Double Diamond model. It consists of four main phases, requiring both convergent and divergent thinking. Additionally, the model considers future research and potential iterations.
In the first two phases of Discovery and Definition, in order to gather knowledge about the current state of the field, I started exploratory research with literature reviews and technology explorations, defining a research focus at the same time and learning about existing design examples. Consequently, I analyzed insights from each method to frame opportunities for the design concept and identify key directions.
Whereas, in the Developing and Delivering phases, I used sketching for idea generation, documentation, and reflection. Sketches served as a base and inspiration for prototype development. In order to develop an interactive artifact, I used digital prototyping, keeping in mind Houde & Hill (1997) approach. During this stage, I considered the key findings from the exploratory phase – literature, interviews, and observations.

4.1 Discovery phase

In the discovery phase, practices already existing in the field were analyzed and further literature research brought up problematic approaches to addictive use of such media. I have learned about the problem and researched about feasible challenges to be solved

4.1.1 Literature reviews

Studying relevant literature helped me to outline the right research approach, trying to focus in this way on feasible design solutions. A literature review was conducted to explore relevant theories and academic articles on technology addiction and interaction design projects to face persuasive technology. As primary search engines in conducting the review I used the ACM Digital Library portal, Google Scholar, ResearchGate, consulting related-field books as well. As a result of the literature review, I gained knowledge in two main fields. First, the representation of the field of study, and then an overview of already existing past design solutions, intended to solve technology addiction. This led me to reflect on the fact that digital addiction can be limited by changing the experience of use of the same technology, in order for the users to benefit from changes of habits.

4.1.2 Semi-structured interviews

Semi-structured interviews were performed to understand digital habits of users throughout the day, in particular with a focus on social media and the tendency of using such technology in the evening or night time, as well as the ways in which this may affect sleeping patterns.
A semi-structured interview is a verbal exchange in which one person, the interviewer, tries to obtain information from another person by asking questions. Although the interviewer prepares a list of predetermined questions, semi-structured interviews are conducted in a conversational manner, giving participants the opportunity to explore issues they feel are important. Those preliminary interviews were conducted on a representative sample of students in Malmö University. The range of hours of investigations was about lunchtime and in evenings. Fieldwork has been done in a three-days-time. Interviews' outcomes demonstrated an overall tendency to scroll social media pages for too long during night-time, before going to bed. This main result was taken into account in designing the next step of quantitative data seeking.

4.1.3 Questionnaire

A questionnaire was formulated to understand how a representative sample of users behaved technologically and digitally and whether they were aware of their digital behaviors. The questionnaire was realized using Google Forms, reaching in this way 17 respondents from different countries. Questionnaires were sent to the research sample during a two-weeks time. The questionnaire was designed to gain several information about the respondent, trying to face some interesting quantitative events needed to be analyzed.

As a general trend result, insights showed the preeminence of Instagram users (88,2%) upon other SnT services. People not using such services explain their choice because they don't like Social media (40%). FoMo (Fear of missing out), a phenomenon leading to hyperconnectivity stress, has been experienced by 52,9% of the sample. Another interesting outcome of the research concerned the time respondents usually check phone notifications. 62,5% of the sample answered that they constantly check phone notifications, aside from the moment of the day they are in. Lastly, the majority of the sample of respondents answered they were not using

any app for monitoring digital wellness (84,6%). The following figures (fig. 15-20) represent some of the questions extracted from the questionnaire used to gather data.

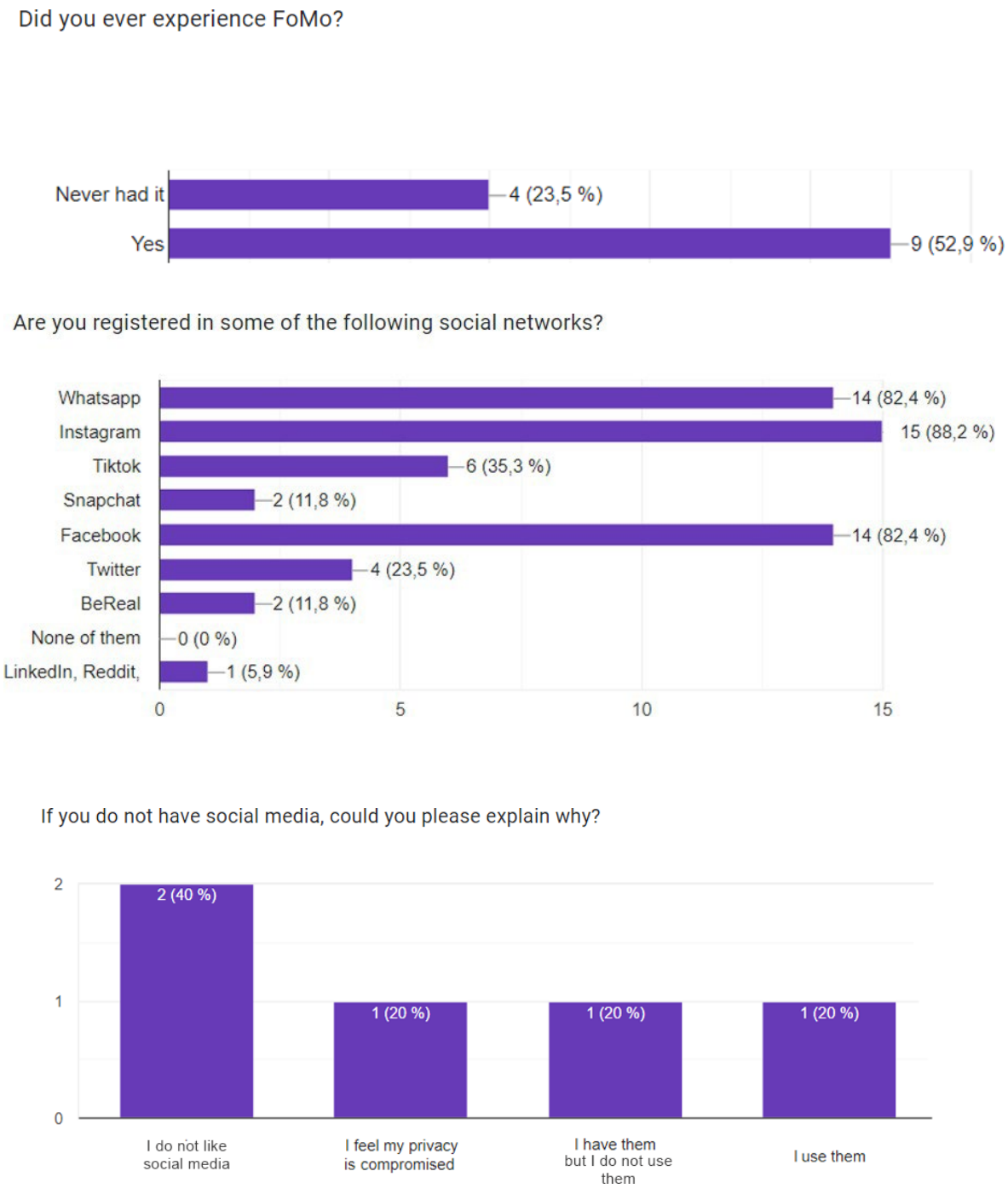
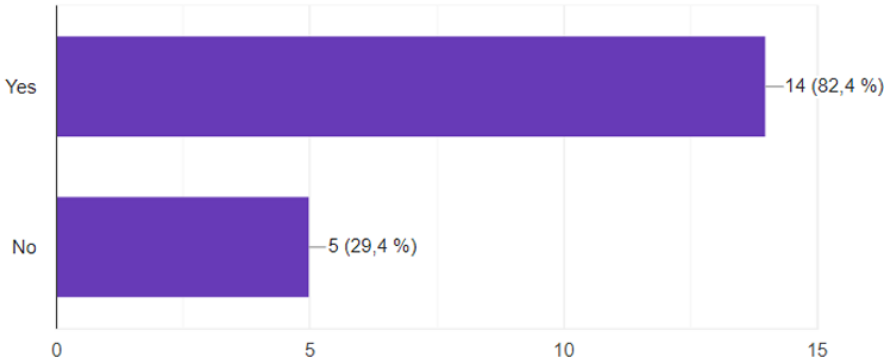
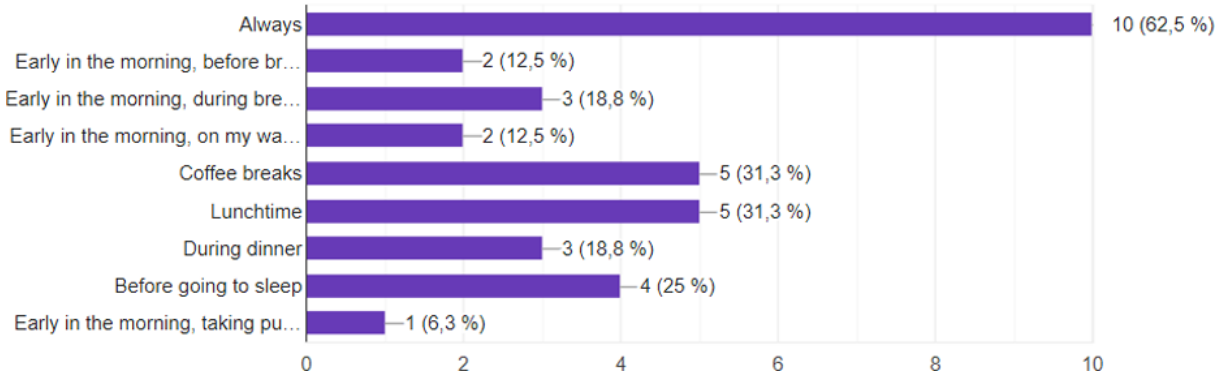


Figure 15, 16, 17 - Questionnaire insights

If you could choose, would you like to reduce the spent time on your digital devices?



In which moments of your day do you usally check your phone notifications?



Do you use any of the following apps for digital wellness?

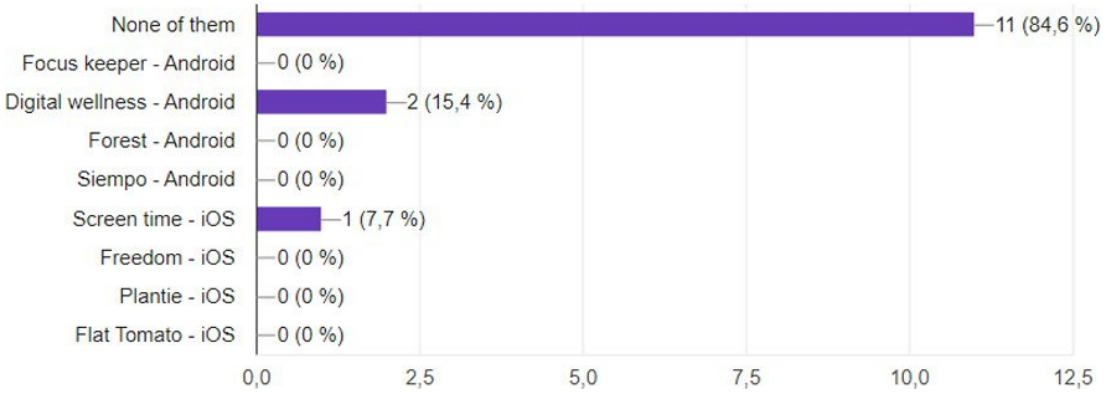


Figure 18, 19, 20 - Questionnaire insights

4.2 Define phase

In the define phase, the information gathered during the discovery phase are organized. Observations are deeply analyzed to define the main issues that have been identified up to this point. Problem definition and problem statement are done in a human-centered way. The research question itself was further analyzed. Night-time use of social media applications affects the ultimate well-being of the user himself in the end. Several ways to overcome this situation were proposed, so that users would actually benefit from such situations. This thesis is guided by a research question "How can Interaction Design support user experience by re-imagining user interaction with technology?"

4.2.1 Affinity diagrams

An activity of affinity diagramming (fig. 21) has been performed to give order to the overall results collected from both semi-structured interviews and a questionnaire form. It was an important phase of the design process, which helped in organizing the gathered data and to give further insights. The term Affinity Diagram (AD) is defined as a technique or a tool that is used in contextual design and in other business practices that organize, sort, analyze and group users' ideas. Insights from both interviews and the questionnaire were organized in order to give a structure to the activities. Affinity diagram organizes information gathered during the discovery phase. Main observations have been better defined in order to identify the majority of problems that have been identified up to this point. Problem definition and problem statement are done in a human-centered manner. Resulting from affinity diagram investigation, insights have been grouped into themes and ranked according to the research question.



Figure 21 - Affinity diagramming

This activity of affinity diagramming helped in defining what kind of design intervention could have been done. Resulting from affinity diagram analysis, it was clear how people experienced bad sleeping habits and patterns after night-time use of digital devices. Most of them would have liked to have something helping their digital habits be healthier.

4.2.2 Early prototyping

An early prototyping activity (fig. 22) has been done to evaluate the work done so far and to speculate about the possible outcomes of the research. This stage of the design process helped me to further understand and narrow down my research focus. Since designing user interfaces would not have helped me innovate this research field, I decided to focus on physical interactive modalities, trying to imagine an interactive design tool, human-centered and focused on user experience. The aim of this first phase of product design reflection is to examine design problems and evaluate feasible solutions.

A prototype is an artifact that approximates a feature (or multiple features) of a product, service, or system (Otto & Wood 2001). A potential version of a digital tool has been created to help users spending too much time on smartphones. The ideas behind this first prototype concept was to enhance well-being, while using digital devices. The user could monitor the time spent on most used apps, like social media, until the phone began to obscure. The graphical interface is intentionally gray, because it does not have to be persuasive and aesthetically appealing to the user. Color plays an important role within a variety of marketing functions.



Figure 22 - Early prototyping stage

4.2.3 How-might-we

In an early stage of the design process, the how-might-we focus activity questions were the following and represented the research questions as well.

> "How might Interaction Design support human wellness through mobile user experiences?"
> and two sub-questions:
> 1. "How much awareness do users have in the unconscious use of digital devices?"
> 2. "How might Interaction designers gain greater insight on the relationship between users' emotions and digital interaction, by expanding their processes and perspectives?"

Later on, in the project itself, I decided to formulate again both the research question and the how-might-we

4.3 Iteration

The iteration itself took place when I refined the prototype, turning it into an interactive experience, enabling in this way users to deal with it. To gain insights about its user experience, I conducted a user testing as a part of an iteration. The test was first designed in the form of a prototype and asking questions to the participants, in the form of a semi-structured interview, conducted in the University Campus of Malmö, involving about 10 people.

4.3.1 Preliminary user testing

The preliminary user testing was performed using a laptop and changing three different layers of the built-in trackpad scrolling sensitivity. Letting users experience the interactive test (fig. 23), and asking them questions about their usage afterwards, led me to achieve how the main results emerging from the test mainly related to users interacting at a slow scrolling screen rate. I provided the participants with scrollable digital interfaces of a social media app (Twitter), to be used at three different speed velocities: slow, intermediate and fast.

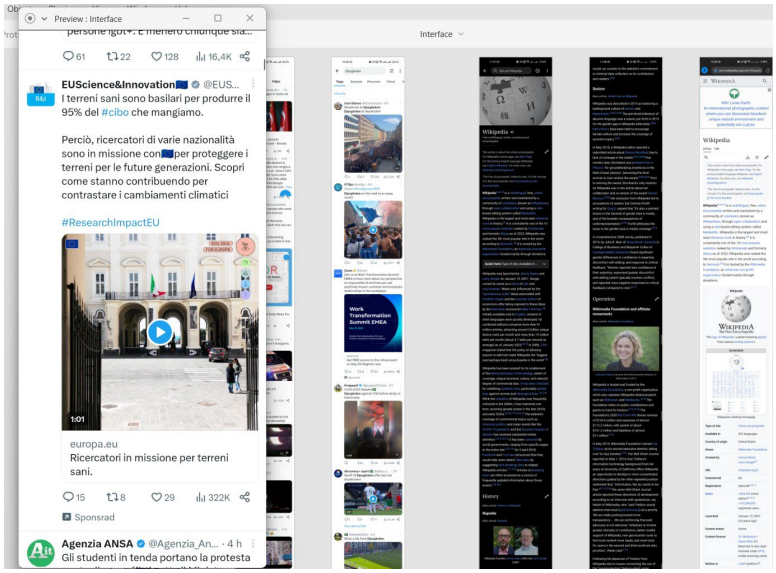


Figure 23 - User testing interfaces

The sample was selected casually in the University Campus (in respect of GDPR regulations). Both textual and social media-like home page scrolling experiences were presented. The social media used in this second semi structured interview was a Twitter long screenshot. As a result, slowest versions of the prototype were the most successful ones, since the majority of the participants answered it would have provoked them to quit the phone and start doing something else instead.

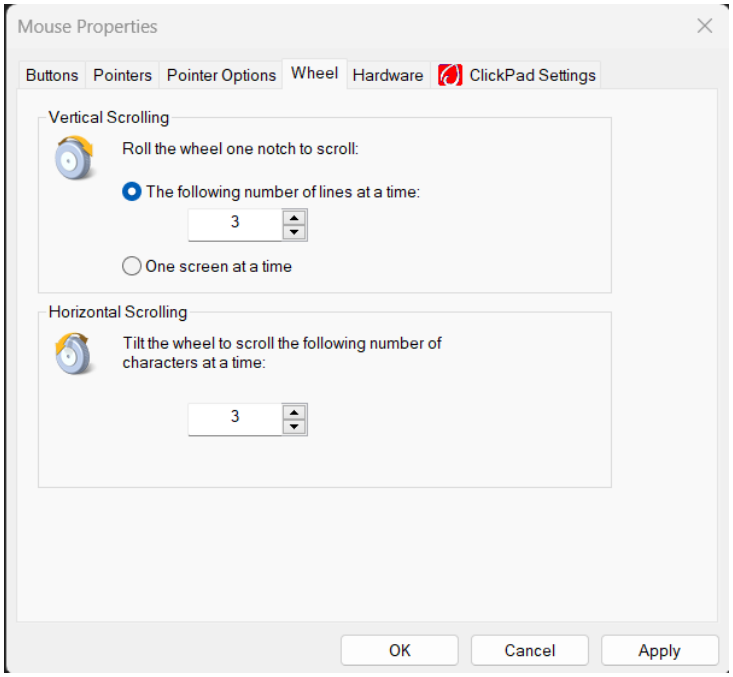


Figure 24 - Changing the built-in trackpad sensitivity on a laptop

4.4 Develop phase

In this phase I analyzed the results from the previous phase, trying to think of possible solutions to eventually be developed as a design proposal. At this stage of the research process the problem has already been defined. Feasible solutions and concepts are then created and enacted.

4.4.1 Sketching

As Buxton writes in Sketching User Experiences (2007), referring to the activity of sketching itself: "It illustrated that the activity of sketching could be extended to other forms than just pencil on paper... sketching as I mean it has more to do with exercising the imagination and understanding (mental and experiential) than about the materials used. Hence, one might use pencil on paper, but one might also use a jar lid, a stick, and a piece of plasticine. It may even involve a computer... The importance of sketching is in the activity, not the resulting artifact (the sketch). If sketches can take on physical form, be they 3D or sculptural, perhaps they can take on even more extended forms that will help us in our quest".

The sketching session began trying to aesthetically figure out catchy and appealing digital interfaces, trying to be captivating but at the same time neutral. During the sketching activity, I explored several ideas, later to be summarized into three main ideas on how to bring users to quit the phone and stop scrolling graphical interfaces.

1. The first idea was to combine three different speed layers of visualization, that the user could compare, investigating the feelings about experiencing them.
2. The second idea was to combine two social contents (written contents and graphical contents), chosen on random topics.
3. The third idea was to combine graphical and color changes of the user interface.

After having explored those three initial ideas during the sketching process, I had to receive

feedback from the sample. It felt important to include the user responses at an early stage so I could evaluate feedback that would be useful in the next phase. By developing the sketches into prototypes, I evaluated my ideas with potential users.

4.5 Deliver phase

In this last phase, the resulting proposal is finalized based on the insights gathered in the previous phases, generating six bodystorming concepts to be roleplayed and enacted. The aim of this activity is finding solutions and seeing if they are adequate to solve the problem stated in the research question.

4.5.1 Bodystorming concepts

The preliminary user testing gave me the idea of enacting six different bodystorming concepts that would ideally help a user in disconnecting from his/her own personal digital device. Based on bodystorming, embodied sketching involves improvisation in the design of experiences supported by technology rather than the technology itself. Improvisation is also central to embodied design improvisation, a multi-stage generative and analytical design process that makes designers' tacit understanding of the design space explicit to other stakeholders in the design process. (Porfirio et al., 2019)

The design method of bodystorming (Brian K Smith. 2014) is used to support interdisciplinary designers with varying formal design experience exploring and supporting user interactions. Although these designers do not have a formal understanding of the research theme through their training or reading of the research literature, they instinctively navigate research topics on a day-to-day basis. Bodystorming method leverages this instinctive ability and involves members of design teams "acting out" or "role-playing" how they expect the design interaction to flow.

Using the body in the interaction design process allows the designer to take advantage of the interconnection between body and mind, to incorporate human kinesthesia into the final product. It also enhances collaboration with other designers (Scott R Klemmer, et al.). Body engaging in bodystorming is also a reflective process that enables designers to consider the nuances of what behaviors might be appropriate for a given scenario. Bodystorming is one such design technique in which designers brainstorm by situating their bodies in the context of the interaction being designed in order to gain insight into the user experience (Colin Burns et al., 1995).

Therefore, I imagined different ways of interacting with smartphone devices.
The first user interface proposal consists in the need of gradually starting to squeeze tighter the smartphone if it wanted to be used any further (fig. 25); subtle hand vibrations would be detected by the accelerometer and gyroscope and an algorithm would analyze and compute the amount of strength of the hand holding the phone. The screen of the phone itself would start flashing so that the user understands he's been using the digital device for too much time.
Users would perceive the squeezing gesture as a way to gradually disengage since having to perform it regularly would entail mental and physical effort. Such kind of effort is not usual throughout the usage session of an interactive device. The expected outcomes of this speculative design intervention would be adding a layer of friction in the normal usage session, so that the user would lose interest in further continuing the same session.

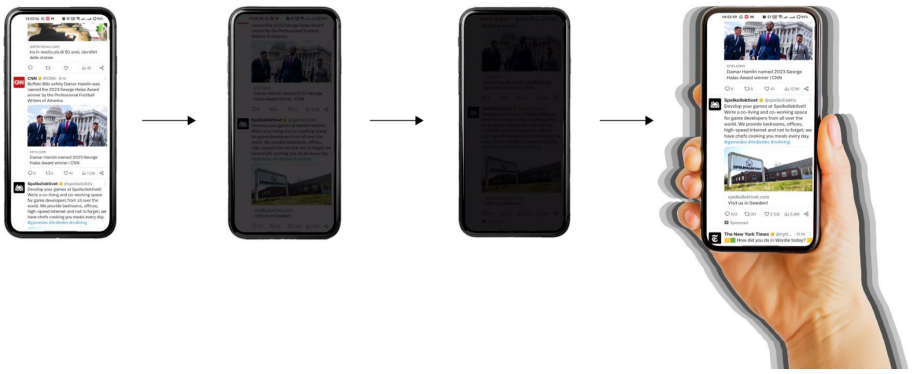


Figure 25 - Gradually squeezing the smartphone

The second one (fig. 26) concerns tilting and shaking the phone to reset normal responsiveness again; the interface would start slowing down and lagging, lowering touch sensitivity. Performing multi-touch gestures would be difficult, and the only way to use the device any further would be interacting with it either by shaking or tilting it.
Performing such modality of use would gradually help users to disengage, as changing the accessibility in this way, introducing a feature of normal state reset would affect the normal fluidity of use. Having to shake the device regularly may entail the users losing interest in further continuing the use session. As a matter of fact, the system would be less responsive, slowing down both the responsiveness and the interactivity, and continuing the usage session would be harder every time, because the whole user experience would revolve around the normal state reset, done through the shake and tilt gestures.



Figure 26 - Shaking the smartphone

The third concept (fig. 27) is about the interface of the phone itself starting to gradually zoom in when in proximity of the user's face and zoom out if distanced from the user himself, having to distance it if wanted it to be used any further; such modality would cause uncomfortableness in

the normal use session and would most likely bring the user to disengage in the end. In this speculative idea of intervention, a normal session of use would be particularly hard, as keeping the phone too close would entail the interface being zoomed too much to be usable. On the other hand, if the phone would be distanced, the interface would then be zoomed out, so that using it would equally be particularly tricky. In both cases, the user experience, being it reading, viewing media, or performing multi-touch gestures would be impractical, and the only way to keep on using the smartphone would be finding the right distance, something not easily achievable.



Figure 27 - Distancing the smartphone

The fourth concept (fig. 28) involves the UI (user interface) slowing down, the scrolling activity gradually lagging, and the need of further pressure input to use the smartphone at a normal rate of responsiveness; the experience of use would be disturbing for users, forcing them to impose more pressure through their fingers when handling the device, so that a normal user experience would be compromised. During a normal usage session, the user would unexpectedly find himself having to physically apply further pressure in order for the user input to be digitally processed.

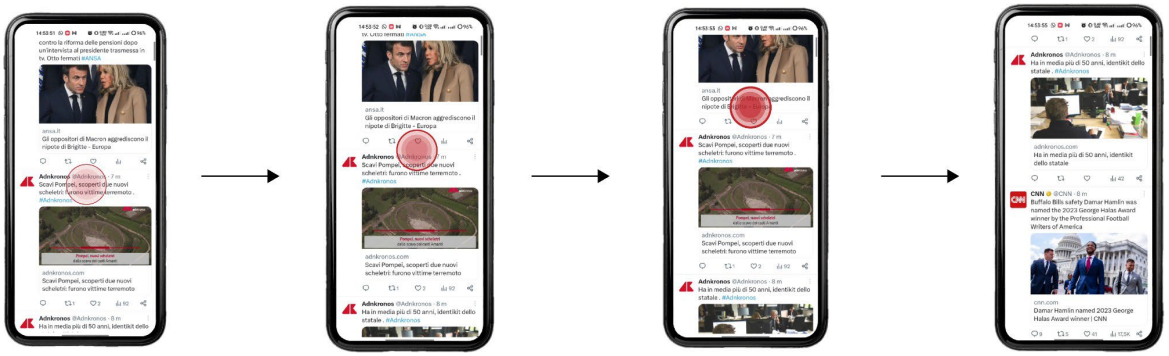


Figure 28 - Inputting further pressure on a smartphone

The fifth concept (fig. 29) consists in a graphically re-arranging UI. Buttons and interactions would be moved in hard-to-reach areas according to the following schematic analysis (fig.28). The figure visually explains how finger movements are performed while handling a smartphone. Red areas represent the hardest level of finger stretching across a smartphone screen, whereas the green ones are the areas of easier reachability. Therefore, while normally using a smartphone, I imagined it suddenly re-arranging the screen position of the icons, main buttons and interactive parts in the UI itself. Continuing the use session would gradually become harder, at least while handling the phone with one hand, since the user experience would be too challenging and definitely less attractive or usable.

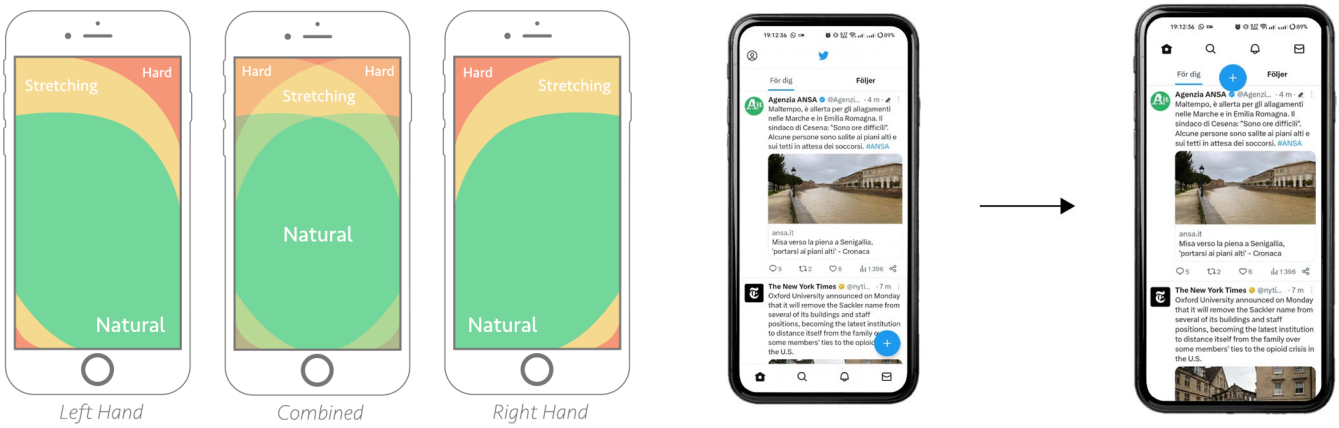


Figure 29 - UI rearranging graphically

The concept of this last interactive proposal (fig. 30) was conceived as having the experience of use altered through the main human perceptive channels, through the alteration of visual, sonic and haptic feedback. In particular, this speculative concept involves color changes and brightness shifts happening on the device, as well as sonic feedbacks and vibration patterns advising the user, after a preset time limit of usage has been reached. Such layers of friction would affect the overall usability of a smartphone's normal user experience. Consequently, I have thought about giving the user agency of either choosing to snooze for 5, 10 or 15 minutes the friction-adding modalities, as well as the possibility of quitting the overused service or turning the smartphone off instead. After a first analysis, it has been evident that by giving the users agency and the possibility to interact with a specific visual UI, either to snooze or stop the friction-adding modalities, would not innovate the literature field. As a matter of fact, similar solutions have already been designed and implemented in already existing applications and digital services. Therefore, the proposal has been resized by removing the user agency interface, so that the user experience alterations would consist in brightness and color shifts, along with sonic and vibration patterns. This kind of adding-friction method would help the user disengage with addictive loops of usage of certain SnT services.

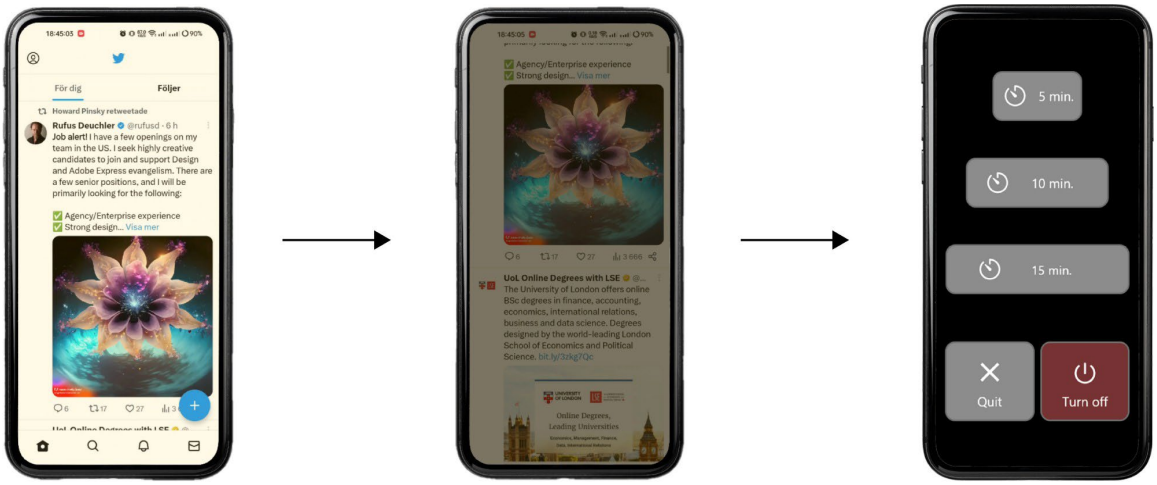


Figure 30 - Concept having multiple friction-adding methods (visual, sonic, haptic feedbacks)

5. Discussion

The research question originating the design process is the following: "How can the aesthetics of interaction be used to give feedback on time spent consuming social media? ". Aside from particular configuration of form, interaction, and functionality, by aesthetics of interaction are intended approaches focused on the experience directly created through the interaction with a product. Product-centered views are distinguished from notions of experience-centered design, urging design to create positive and meaningful experiences through technology, but not to make the technology into an experience in itself (Lenz. et al, 2014).
The main idea of this thesis is to debate how it would be possible to have different types of feedback of SnT services over usage in digital mobile devices. Drawing examples from a scientific literature background outlining health and social problems of overuse and merging it with an interaction design one, a speculative approach of giving overuse feedback through different interactive modalities is proposed. Is it possible to break away the status quo provided by the companies behind the already existing digital wellbeing applications? For example, why are most of them still relying on the same control patterns (e.g. built-in system clock check, usage charts to cause a sense of guilt in the user, time-limiting features in the use of digital apps, etc.) instead of having more possible aesthetic nuances of interaction?

Aesthetic interaction is defined through the process of use, as "products that are beautiful in use". (Djajadiningrat et al 2004). This is the most simple definition with just one single word "beauty" behind the aesthetic value. There are no further explanations about stimuli, interpretations, or senses. The usability of a product is tightly tied to aesthetics of appearance that may increase the usability, but good usability can also contribute to perceived aesthetics of interaction. This simple definition is easy to understand but modeling the aesthetics of the interface requires a more explicit definition. Aesthetics of interaction is a set of principles concerned with the nature and appreciation of beauty of interactive products. Aesthetic value also acts as a quality dimension of user experience together with usability and pleasure of use (Hassenzahl 2011).

It is well known about the importance of technology in our everyday life, since the ease of being world-wide connected to everyone brings many benefits. Greater and easier access to information and services has undoubtedly been allowed by technology, enabling and facilitating social networking, openness and collaboration. On the other hand, the negative overuse downsides have been widely stated during this thesis, identifying them both in health and wellbeing consequences, as the awareness of the potential disadvantages of its communicative ways are growing. Kraut et al. (1998) found out how greater use of the internet was associated with declines in family communication, decreases and distancing in social circles, and increases in depression and loneliness. It might end up reducing face-to-face contact, to the point that weaker social relationships replace stronger ones.

As a design proposal, six different bodystorming concepts were proposed in this thesis to provide an answer to the research question, because using human sense can increase user awareness of bad technological habits. Due to the bodystorming concepts' speculative nature, there is no certainty about their intended social outcomes. It would be possible to argue that such interventions would not bring the user to lower the phone usage amount of time or to cause better digital habits. The general level of friction provided by all the concepts would cause the usage session to be negatively affected in the ease of use.

The main aim behind the concepts has been to prevent addictive loops from being constantly connected to SnT services and smartphones in general.
The speculative concepts aim to make the users feel motivated to engage in different activities or stop using social media, challenging themselves, their digital habits and their well-being and healthy habits. Results from an intermediate phase of the design process, showed that users testing a prototype having three different responsivity levels in the scrolling experience, were led to quit social media after having experienced the slowest scrolling interface. This outcome led me to design the six speculative concepts already described, intending to help users to reach their own time-management purposes, since the test revealed that the sense of time passing is altered by the interface ease of use. Using human perceptive channels in order to interfere with the user experience of technological devices has been important in reaching the thesis own purposes. Users, as a matter of fact, after having experienced design products, felt they no longer wanted to spend more time with their own smartphones.

6. Conclusion

By trying to answer to the research question of how the interaction aesthetics could be used to give feedback on spent digital time on social media, I found out that altering the interactive modalities of the user experience, once the preset daily time limit of use had been reached, proved to be a valid strategy to help users disengage, bringing them to eventually perform healthier natural behaviors instead.

The psychological outcomes of addictive digital user experiences made me speculate on a design proposal, giving users feedback about the excessive time they might be spending on mobile technology, using features that progressively reduce accessibility and consequently addictiveness. In this project, I explored opportunities about how interaction design aesthetics might support the user experience of digital devices and give feedback on time spent consuming

social media content. The project's objective is to disincentivize over usage through making the users aware of the digital time spent, leading them to eventually engage in alternative activities instead.

I investigated this problem through ethnographic methods such as semi-structured interviews, a questionnaire and a user-testing. The main outcomes from this type of quantitative and qualitative research methods concerned ways to enhance the awareness of spending too much time in front of digital devices. From the studies emerged that the best way of dealing with the issue was through the use of friction in the scrolling interfaces, so I started the design process by unfolding the research question and organizing the gathered insights. After an affinity diagramming activity, I designed a user test supposed to understand the scrolling habits and the related levels of awareness in users. By comparing three different speed rates of scrolling consumption, I found out that people perceive time and their content consumption differently, according to interfaces' responsivity rate. Consequently, in a speculative approach, I supposed that slowed down interfaces are the ones that will most likely help users in disengaging from the smartphone. I created six different speculative concepts trying to trigger different digital behaviors in users. The outcome suggests that such digital design concepts would meet the requirements necessary to answer the research question.

Living in an overconnected era, sometimes too immersed in social media usage to perceive digital overuse as normal, may cause negative social consequences that would eventually require further behavioral changes, in order to be broke down and solved.
All the presented solutions were conceived in an attempt to help users control or limit smartphone use, causing a behavioral change, trying to limit short-term digital gratification and giving the possibility of achieving long-term real-life goals instead.
These speculative concepts would ideally help reflecting upon the eventual excessive use and limiting overuse. In each situation, certain cues (e.g. ringtone, urgency, escape) trigger reflective process leading to disengagement and developing of conscious usage rules to regulate smartphone use. Non-use or very infrequent use of smartphone does not have to be necessarily reflective, meaning that sometimes users tend to unconsciously pick-up devices and open social media apps just because they developed the habit. These types of behavioral dynamics are the ones supposed to happen in case of compulsive or excessive use. Using a smartphone or a social media mobile app for an excessive amount of time share the same state of unconsciousness as picking repeatedly the device in order not to miss out for notification.
A decision to disengage from smartphones occurs on two levels: logic of appropriateness and logic of consequence. The first entails the perception others may have of the smartphone overuser and the context of use, whereas the second concerns the physical or social consequences this may cause. The six concepts aim to lower levels of digital immersion, presence or absorption which may cause addictiveness. Some of them (concept 1-4) are more suited to reset a normal level of the device responsivity other than just preventing overuse (concept 5-6). Each of them has its own level of functionality, limiting the user's operability and eventually driving him not to engage anymore with SnT in the very moment.

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Appendix

Semi structured interview questions:

- In which moment of your day do you find yourself using social media more often?
- If in the evening, how long?
- How many times in a week?
- In a normal week, how many evenings would you find yourself using social media? Is it in the evenings, at night or before going to sleep?
- Is the use of social media relaxing yourself before going to sleep?
- If you had the possibility to have a built-in system notifying you the excessive use of a certain social media, would that bother you?
- In which way would you prefer to be notified about this?
- Would that be through sonic, visual or haptic feedback?
- How would that work?

User testing questions

Questions	1st participant	2nd participant	3rd participant	4th participant	5th participant	6th participant
How long do you think that took? SLOW	5 min / 4 min	4 min / 5 min	4.30 min / 4 min	5 min / 4 min	4 min / 5 min	4 min / 5 min
	3 min / 2 min	3.30 min / 2 min	4 min / 2.30 min	2 min / 3 min	4 min / 2 min	3.30 min / 2 min
FAST	3 min / 2 min	2 min / 3 min	3 min / 2 min	3 min / 2 min	3 min / 2.30 min	2 min / 3 min
How many images did you see on the feed?	100 120 140	80 100 120	50 80 90	65 90 100	70 80 100	80 90 100
Has your level of awareness alway been on the same level?	No, at a slow rate I was barely reading and trying to understand	No	No, because while I was reading I started thinking about something else	No, at the beginning my attention was high but later on it dropped	No, because my attention was taken from an image	No, I couldn't follow the slowest one
Which differences did you notice between the three different sensitivities of scrolling?	My mental tiredness	My visual fatigue	My level of awareness and understanding of text and images	My willingness to use the smartphone any longer	The tolerability of usage	The speed levels
Which one of the three different levels of sensitivity did you prefer?	The intermediate one is a normal speed rate	Slowest, because you're more aware	Fast, because it's the speed i'm used to	I don't like the slowest because it's too slow	The slowest made me do something else	The slowest one made me bored
How did the slowest level affect the digital experience?	The experience of use barely became usable	I started getting very annoyed	It remembered me that I have been using it for too much time	It makes me think other things	It makes me lose the attention of what is on the screen	I would like to shut the phone down
In the textual interfaces how much time do you think you spent reading?	2 min	A lot	3 min	There was too much text and I did not want to read	4 min	3 min
Which friction method would help you disengage?	Zooming in/ Zooming out	UI graphically rearranges	Color changes	Zooming in/ zooming out	UI slows down and you need to input further pressure	The slowest one for sure

Questions	7th participant	8th participant	9th participant	10th participant
How long do you think that took?	5 min / 4 min 3 min / 2 min 3 min / 2 min	4.30 min / 3 min 3 min / 2.30 min 2.30 min / 2 min	4 min / 5 min 3 min / 2 min 1.30 min / 2 min	5 min / 4.30 min 3 min / 2 min 3 min / 2 min
How many images did you see on the feed?	160 200 220	160 200 220	160 200 220	160 200 220
Has your level of awareness alway been on the same level?	No, I was tired of watching the interface and scrolling at the same time	Yes, on the slowest I was able to read and focus on everything	No, I found the slowest one very annoying due to the delay	No, on the intermadiate I was more kneen on following
Which differences did you notice between the three different sensitivities of scrolling?	The speed levels	My attention	My short-term memory	My perception of time
Which one of the three different levels of sensitivity did you prefer?	The intermediate one	The slowest	The fastest	The slowest
How did the slowest level affect the digital experience?	In the night time it remembers me I have to go to sleep	It makes me think about how useless is the time spent in this way	I started to sense the need of doing something else	The experience of use barely became usable
In the textual interfaces how much time do you think you spent reading?	When the interface was slow I was able to read the content in a better way	The fastest one couldn't make me see anything clearly	The fastest one didn't really get me the overall sense of it	The intermediate one let me read only the main and bold titles
Which friction method would help you disengage? (1-6)	Squeezing the phone	Tilting and shaking the phone	Slowing down the UI	Distancing the phone